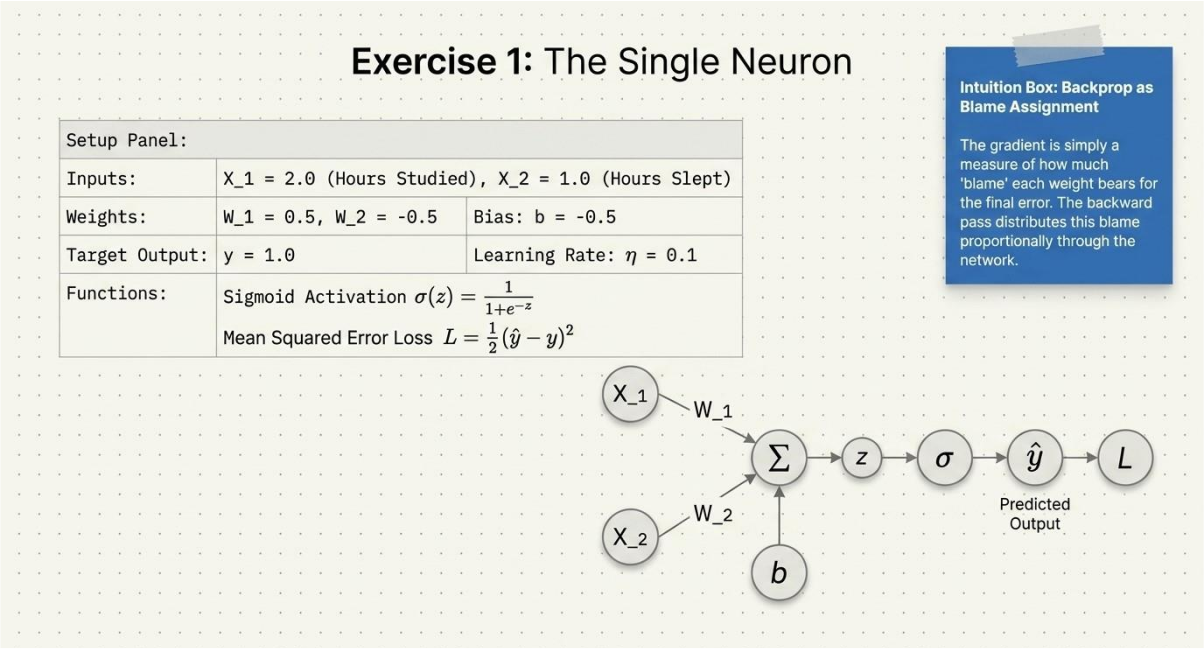
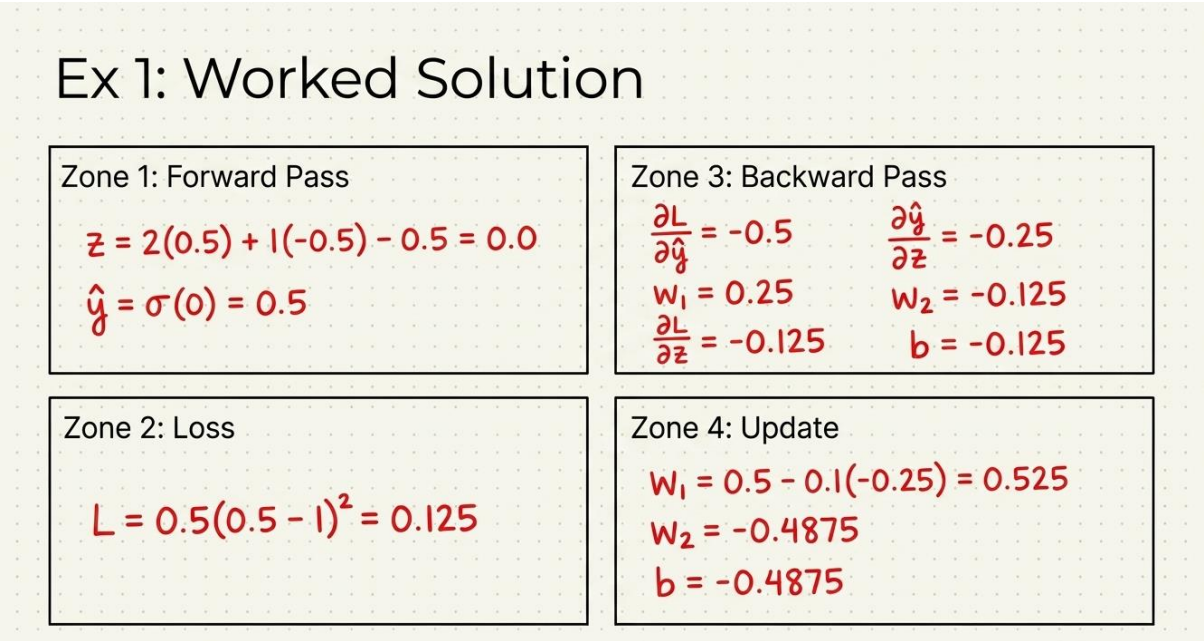
FIT3181-Week 4: Additional exercises' Calculation steps:

Exercises 1: The Single Neuron





Zone 1:

Step 1: Forward pass

The neuron first computes a weighted sum plus bias:

$$z = x_1 w_1 + x_2 w_2 + b$$

Insert the given values:

$$z = 2(0.5) + 1(-0.5) - 0.5$$

Multiply each input by its weight:

$$z = 1 - 0.5 - 0.5$$

Combine the terms:

$$z = 0$$

Now use the **sigmoid activation**:

$$\hat{y} = \sigma(z)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Substitute $z = 0$:

$$\hat{y} = \frac{1}{1 + e^{0}}$$

$$\hat{y} = \frac{1}{2}$$

$$\hat{y} = 0.5$$

Zone 2: Loss:

The squared-error loss is:

$$L = \frac{1}{2}(\hat{y} - y)^2$$

Substitute prediction 0.5 and target 1:

$$L = \frac{1}{2}(0.5 - 1)^2$$

$$L = \frac{1}{2}(-0.5)^2$$

$$L = 0.125$$

Zone 3: Backward pass:

Start from the loss and move backward.

The derivative of squared error with respect to prediction is:

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

Substitute the known prediction and target:

$$\frac{\partial L}{\partial \hat{y}} = 0.5 - 1$$

$$\frac{\partial L}{\partial \hat{y}} = -0.5$$

The derivative of sigmoid is:

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y})$$

Substitute $\hat{y} = 0.5$:

$$\frac{\partial \hat{y}}{\partial z} = 0.5(1 - 0.5)$$

$$\frac{\partial \hat{y}}{\partial z} = 0.25$$

Now apply the chain rule:

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z}$$

$$\frac{\partial L}{\partial z} = (-0.5)(0.25)$$

$$\frac{\partial L}{\partial z} = -0.125$$

This is the neuron's **delta**: the error signal that will be distributed to its parameters.

Zone 4: Updates:

Because:

$$z = x_1 w_1 + x_2 w_2 + b$$

the local derivatives are:

$$\frac{\partial z}{\partial w_1} = x_1$$

$$\frac{\partial z}{\partial w_2} = x_2$$

$$\frac{\partial z}{\partial b} = 1$$

Apply the chain rule to w_1:

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z}\frac{\partial z}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = (-0.125)(2)$$

$$\frac{\partial L}{\partial w_1} = -0.25$$

For w_2:

$$\frac{\partial L}{\partial w_2} = (-0.125)(1)$$

$$\frac{\partial L}{\partial w_2} = -0.125$$

For bias:

$$\frac{\partial L}{\partial b} = (-0.125)(1)$$

$$\frac{\partial L}{\partial b} = -0.125$$

Gradient-descent update

Use the update rule:

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta\frac{\partial L}{\partial \theta}$$

The learning rate is:

$$\eta = 0.1$$

Update w_1:

$$w_{1,\text{new}} = 0.5 - 0.1(-0.25)$$

$$\boxed{w_{1,\text{new}} = 0.525}$$

Update w_2:

$$w_{2,\text{new}} = -0.5 - 0.1(-0.125)$$

$$\boxed{w_{2,\text{new}} = -0.4875}$$

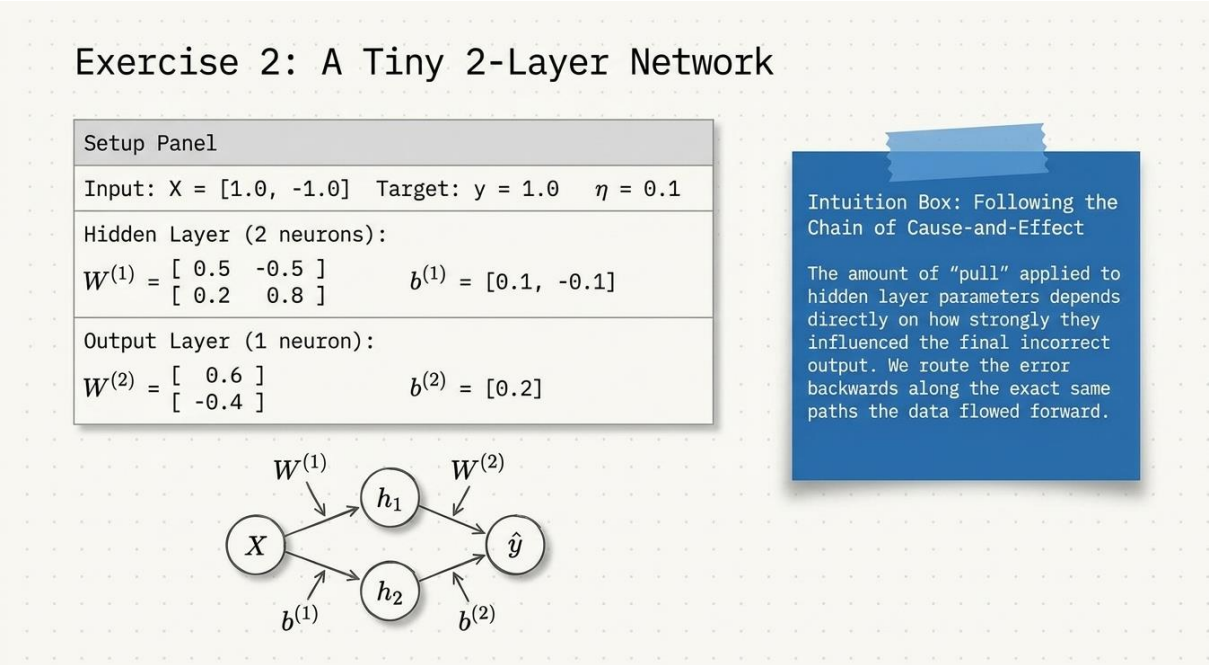
Update b:

$$b_{\text{new}} = -0.5 - 0.1(-0.125)$$

$$\boxed{b_{\text{new}} = -0.4875}$$

Because all gradients are negative, subtracting the gradient increases every parameter.

Exercise 2: A Tiny 2-Layer Network



The network has this direction:

$$X \to z^{(1)} \to h^{(1)} \to z^{(2)} \to \hat{y} \to L$$

The given quantities are:

$$X = [1, -1]$$

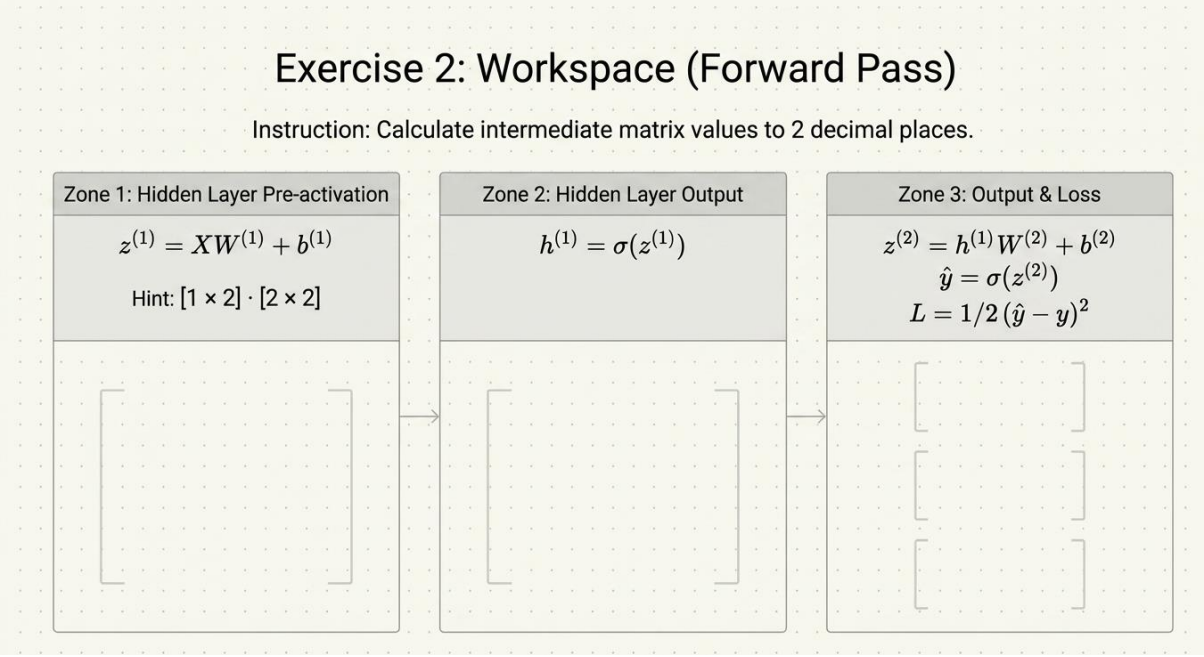
$$W^{(1)} = \begin{bmatrix} 0.5 & -0.5 \\ 0.2 & 0.8 \end{bmatrix}$$

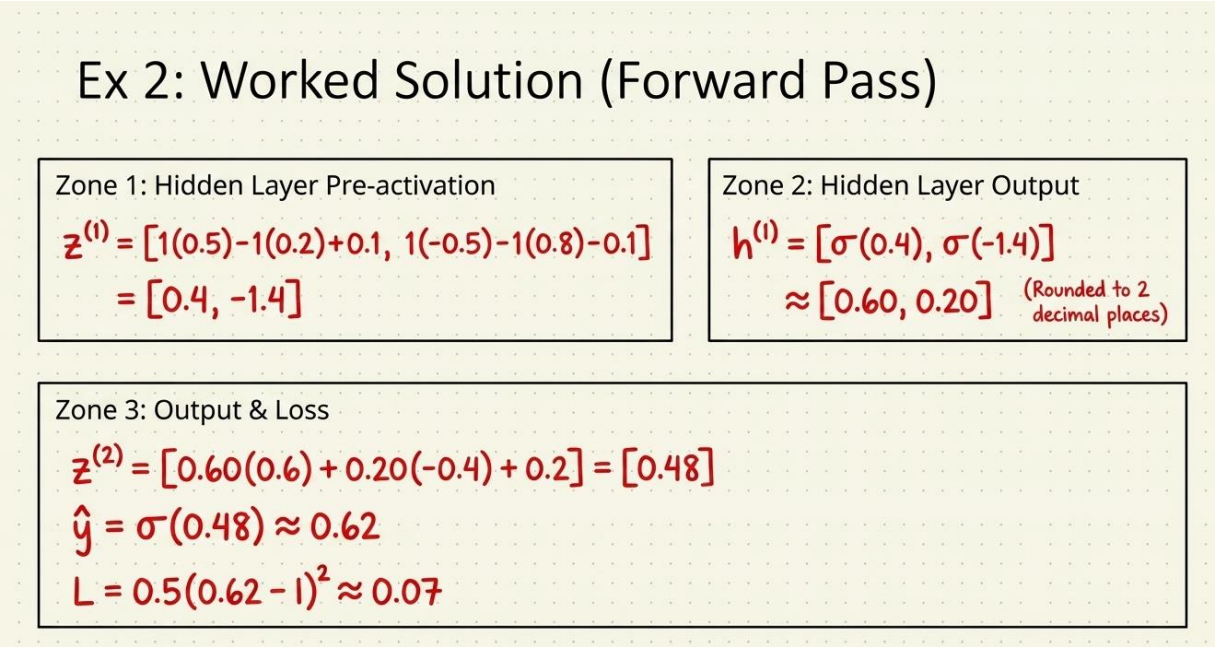
$$b^{(1)} = [0.1, -0.1]$$

$$W^{(2)} = \begin{bmatrix} 0.6 \\ -0.4 \end{bmatrix}$$

$$b^{(2)} = 0.2$$

$$y = 1$$





Zone 1: Hidden pre-activation

Use the hidden-layer linear transformation:

$$z^{(1)} = XW^{(1)} + b^{(1)}$$

Compute the first hidden neuron:

$$z_1^{(1)} = 1(0.5) + (-1)(0.2) + 0.1$$

$$z_1^{(1)} = 0.5 - 0.2 + 0.1$$

$$z_1^{(1)} = 0.4$$

Compute the second hidden neuron:

$$z_2^{(1)} = 1(-0.5) + (-1)(0.8) - 0.1$$

$$z_2^{(1)} = -0.5 - 0.8 - 0.1$$

$$z_2^{(1)} = -1.4$$

Therefore:

$$\boxed{z^{(1)} = [0.4, -1.4]}$$

Zone 2: Hidden-layer output

Apply sigmoid separately to both entries:

$$h^{(1)} = \sigma\left(z^{(1)}\right)$$

$$h_1^{(1)} = \sigma(0.4)$$

$$h_1^{(1)} \approx 0.5987$$

$$h_2^{(1)} = \sigma(-1.4)$$

$$h_2^{(1)} \approx 0.1978$$

Thus:

$$\boxed{h^{(1)} \approx [0.5987, 0.1978]}$$

The slide's solution had simplified these to:

$$[0.60 \quad 0.20]$$

Zone 3: Output and loss:

Use the output layer equation:

$$z^{(2)} = h^{(1)}W^{(2)} + b^{(2)}$$

Using the more precise hidden values:

$$z^{(2)} = 0.5987(0.6) + 0.1978(-0.4) + 0.2$$

$$z^{(2)} = 0.3592 - 0.0791 + 0.2$$

$$z^{(2)} \approx 0.4801$$

Compute the sigmoid prediction:

$$\hat{y} = \sigma\left(z^{(2)}\right)$$

$$\hat{y} = \sigma(0.4801)$$

$$\hat{y} \approx 0.6178$$

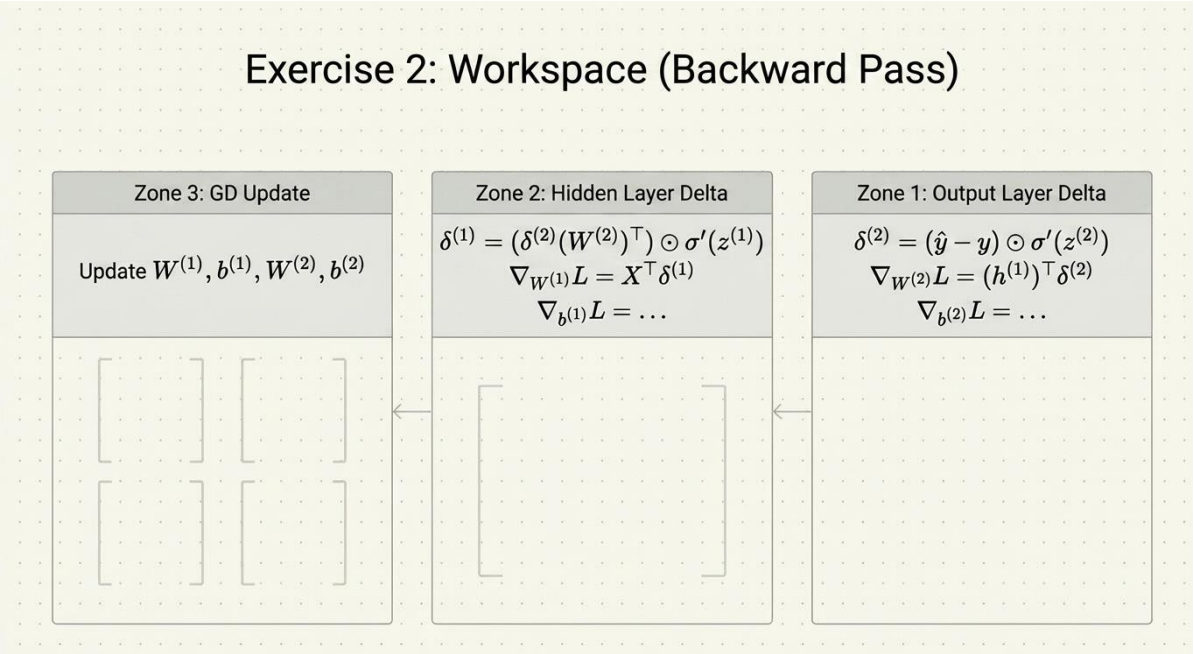
Compute the loss:

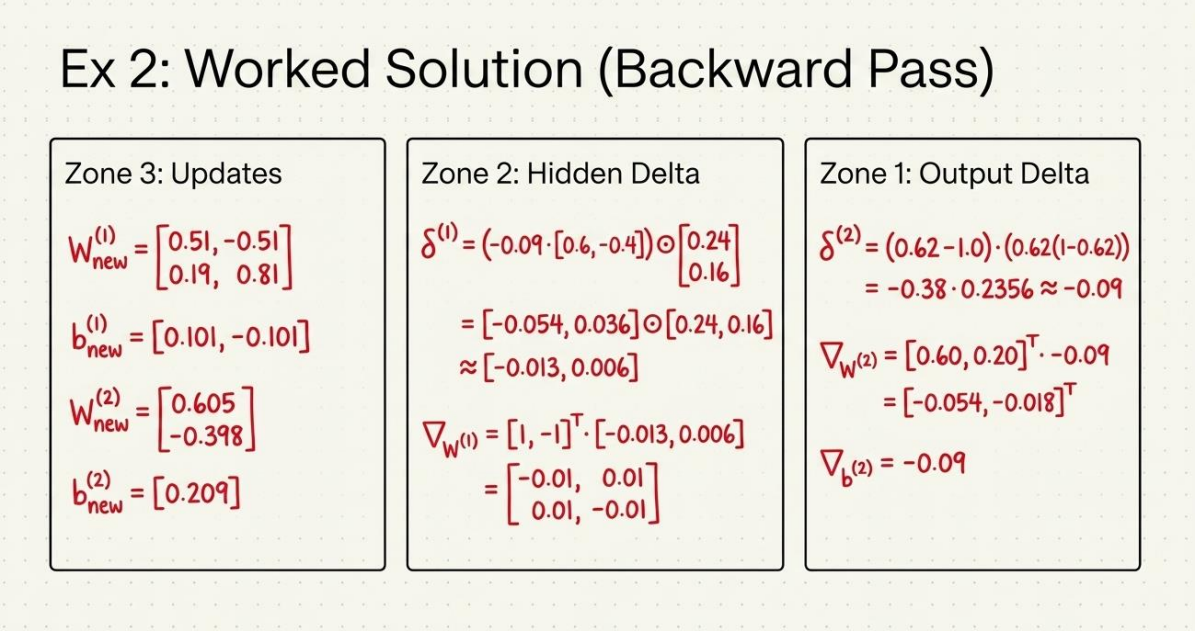
$$L = \frac{1}{2}(\hat{y} - y)^2$$

$$L = \frac{1}{2}(0.6178 - 1)^2$$

$$L = \frac{1}{2}(-0.3822)^2$$

$$\boxed{L \approx 0.0730}$$





Backward Pass: Zone 1:

For sigmoid plus squared error:

$$\delta^{(2)} = \frac{\partial L}{\partial z^{(2)}} = (\hat{y} - y)\sigma'(z^{(2)})$$

The sigmoid derivative is:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

Therefore:

$$\sigma'(z^{(2)}) = \hat{y}(1 - \hat{y})$$

Using the precise prediction:

$$\sigma'(z^{(2)}) = 0.6178(1 - 0.6178)$$

$$\sigma'(z^{(2)}) \approx 0.2361$$

Now multiply the loss error by this derivative:

$$\delta^{(2)} = (0.6178 - 1)(0.2361)$$

$$\delta^{(2)} \approx -0.0902$$

$$\boxed{\delta^{(2)} \approx -0.0902}$$

Zone 1: Output-layer gradients

The weight-gradient formula is:

$$\nabla_{W^{(2)}} L = \left(h^{(1)}\right)^T \delta^{(2)}$$

For the first output weight:

$$\frac{\partial L}{\partial W_1^{(2)}} = 0.5987(-0.0902)$$

$$\frac{\partial L}{\partial W_1^{(2)}} \approx -0.0540$$

For the second output weight:

$$\frac{\partial L}{\partial W_2^{(2)}} = 0.1978(-0.0902)$$

$$\frac{\partial L}{\partial W_2^{(2)}} \approx -0.0178$$

Thus:

$$\boxed{\nabla_{W^{(2)}} L \approx \begin{bmatrix} -0.0540 \\ -0.0178 \end{bmatrix}}$$

The output bias directly adds to $z^{(2)}$, so its local derivative is 1:

$$\nabla_{b^{(2)}} L = \delta^{(2)}$$

$$\boxed{\nabla_{b^{(2)}} L \approx -0.0902}$$

Zone 2: Hidden-layer delta

Here is the most important backpropagation idea. Each hidden neuron affects the loss through its outgoing output-layer weight. First route the output delta backward:

$$\delta^{(2)}\left(W^{(2)}\right)^{\mathsf{T}}$$

$$-0.0902[0.6 \quad -0.4]$$

$$= [-0.0541 \quad 0.0361]$$

Now calculate the derivative of each hidden sigmoid:

$$\sigma'\left(z^{(1)}\right) = h^{(1)} \odot \left(1 - h^{(1)}\right)$$

For hidden neuron 1:

$$0.5987(1 - 0.5987)$$

$$\approx 0.2403$$

For hidden neuron 2:

$$0.1978(1 - 0.1978)$$

$$\approx 0.1587$$

So:

$$\sigma'\left(z^{(1)}\right) \approx [0.2403 \quad 0.1587]$$

Use element-wise multiplication, denoted by $\odot$:

$$\delta^{(1)} = \left(\delta^{(2)}\left(W^{(2)}\right)^{\top}\right) \odot \sigma'\left(z^{(1)}\right)$$

$$\delta^{(1)} = [-0.0541 \quad 0.0361] \odot [\,0.2403 \quad 0.1587]$$

$$\boxed{\delta^{(1)} \approx [-0.0130 \qquad 0.0057]}$$

Hidden-layer gradients

Use the matrix formula:

$$\nabla_{W^{(1)}} L = X^{\top}\delta^{(1)}$$

Since:

$$X^{\top} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

the outer product is:

$$\nabla_{W^{(1)}} L = \begin{bmatrix} 1 \\ -1 \end{bmatrix} [-0.0130 \qquad 0.0057]$$

$$\boxed{\nabla_{W^{(1)}} L \approx \begin{bmatrix} -0.0130 & 0.0057 \\ 0.0130 & -0.0057 \end{bmatrix}}$$

The hidden bias gradient is simply the hidden delta:

$$\nabla_{b^{(1)}} L = \delta^{(1)}$$

$$\boxed{\nabla_{b^{(1)}} L \approx [-0.0130 \qquad 0.0057]}$$

Zone 3: Update every parameter

The intended learning rate is:

$$\eta = 0.1$$

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Prepared by Dr. Sicily Ting

Use:

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_{\theta} L$$

For the output weights:

$$W_{\text{new}}^{(2)} = \begin{bmatrix} 0.6 \\ -0.4 \end{bmatrix} - 0.1 \begin{bmatrix} -0.0540 \\ -0.0178 \end{bmatrix}$$

$$\boxed{W_{\text{new}}^{(2)} \approx \begin{bmatrix} 0.6054 \\ -0.3982 \end{bmatrix}}$$

For the output bias:

$$b_{\text{new}}^{(2)} = 0.2 - 0.1(-0.0902)$$

$$\boxed{b_{\text{new}}^{(2)} \approx 0.2090}$$

For the hidden weights:

$$W_{\text{new}}^{(1)} = \begin{bmatrix} 0.5 & -0.5 \\ 0.2 & 0.8 \end{bmatrix} - 0.1 \begin{bmatrix} -0.0130 & 0.0057 \\ 0.0130 & -0.0057 \end{bmatrix}$$

$$\boxed{W_{\text{new}}^{(1)} \approx \begin{bmatrix} 0.5013 & -0.5006 \\ 0.1987 & 0.8006 \end{bmatrix}}$$

For the hidden biases:

$$b_{\text{new}}^{(1)} = [0.1 \quad -0.1] - 0.1[-0.0130 \qquad 0.0057]$$

$$\boxed{b_{\text{new}}^{(1)} \approx [0.1013 \quad -0.1006]}$$